

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Regular and Supplementary Winter Examination – 2024

Course: B. Tech

Semester : VII

Branch : Electronics & Telecommunication Engineering / Electronics and Communication Engineering/ Electronics Engineering

Subject Code & Name: Fiber Optic Communication / BTETPE702D/BTEXPE702D

Max Marks: 60

Date: 07/02/2025

Duration: 3 Hr

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

	(Level/CO)	Marks
Q. 1 Objective type questions. (Compulsory Question)		12
Which of the following best describes the vector nature of light?	CO1	1
a) Light behaves only as a particle.		
b) Light behaves only as a wave.		
c) Light has both electric and magnetic field components that oscillate perpendicular to each other.		
d) Light is a scalar quantity with no directionality.		
2 What is the primary mechanism by which light propagates in a cylindrical dielectric rod?	CO1	1
a) Reflection from the outer surface.		
b) Total internal reflection at the core-cladding interface.		
c) Absorption and re-emission of light.		
d) Diffraction at the edges of the rod.		
3 Which phenomenon is best explained by the wave model of light?	CO2	1
a) Rectilinear propagation.		
b) Reflection.		
c) Interference and diffraction.		
d) Refraction.		

4	In a step-index fiber, what determines the number of modes supported? a) Core diameter and wavelength of light. b) Cladding thickness. c) Refractive index of the cladding only. d) Attenuation coefficient.	CO3	1
5	What is the main effect of dispersion in optical fibers? a) Increase in signal strength. b) Broadening of optical pulses. c) Reduction in attenuation. d) Increase in bandwidth.	CO3	1
6	Which method is commonly used for the fabrication of optical fibers? a) Chemical vapor deposition (CVD). b) Mechanical drawing. c) Extrusion. d) Electroplating.	CO3	1
7	Which of the following is a key difference between LEDs and Lasers? a) LEDs produce incoherent light, while Lasers produce coherent light. b) LEDs have a narrower spectral width than Lasers. c) Lasers are less efficient than LEDs. d) LEDs require an optical cavity to operate.	CO4	1
8	What is the responsivity of a photo-detector? a) The ratio of output current to input optical power. b) The ratio of output voltage to input optical power. c) The ratio of output optical power to input current. d) The ratio of output optical power to input voltage.	CO4	1
9	What is the main application of directional couplers in optical communication systems? a) Amplification of signals. b) Splitting or combining optical signals. c) Modulating optical signals. d) Generating optical signals.	CO4	1
10	What is the operating principle of a Raman amplifier? a) Stimulated emission. b) Stimulated Raman scattering. c) Spontaneous emission. d) Absorption.	CO4	1
11	What does WDM stand for in optical communication systems? a) Wavelength Division Multiplexing. b) Waveguide Division Multiplexing.	CO4	1

- c) Wideband Division Multiplexing.
- d) Wireless Division Multiplexing.

12	Which of the following is a nonlinear effect in fiber optic links?	CO5	1
	a) Self-phase modulation (SPM).		
	b) Group velocity dispersion (GVD).		
	c) Attenuation.		
	d) Refraction.		

Q. 2 Solve the following.

A)	Explain the following terms:	CO1	12
	(a) Numerical Aperture		6
	(b) Total Internal Reflection		
	(c) Acceptance angle		
B)	Calculate the numerical aperture, fractional refractive index Δ and acceptance angle of a fiber having the refractive index of core is 1.44 and the refractive index of cladding is 1.40.	CO1	6

Q.3 Solve the following.

A)	Derive an expression for number of modes exists in step indexed fiber. Also explain about mode field diameter.	CO1	12
B)	A graded index fiber with parabolic refractive index has $n_1=1.48$ and $n_2=1.46$ if core radius is 20 μm . Find the number of modes at 1300nm and 1550nm.	CO1/CO2	6

Q. 4 Solve Any Two of the following.

A)	Explain the working principles of PIN-diodes.	CO3	12
B)	What is the source of power penalty? Explain	CO3	6
C)	What is threshold condition for LASER oscillation? Explain in detail.	CO3	6

Q.5 Solve Any Two of the following.

A)	Explain the working principles of electro-optic switches and how they are used to control the flow of optical signals.	CO3/CO4	12
B)	Explain the operating principles of Erbium-Doped Fiber Amplifiers (EDFAs) as optical amplifiers.	CO4	6
C)	What are WDM networks? Explain the principle of WDM networks in detail.	CO4	6

Q. 6 Solve Any Two of the following. **12**

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| A) What is the principle behind soliton-based communication in optical fibers? | CO4 | 6 |
| B) Explain the concept of self-phase modulation in detail. | CO5 | 6 |
| C) Explain how solitons are formed and maintained, and why they are significant for long-distance communication. | CO5 | 6 |

*** End ***

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